

(Distillation)

Pre-Lab Plan

Unit Operations Lab 1

10/29/2025

Group 1, Tuesday, AM section

Team Member Names:

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1. Experimental Objective (5 pts)

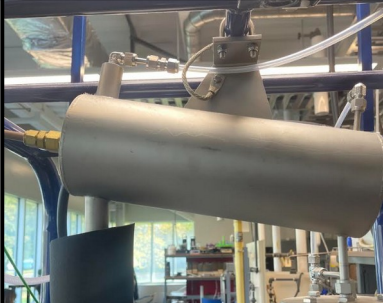
The objective of this experiment is to study the separation characteristics of a binary ethanol–water mixture using a distillation column. In the first part, the effect of reboiler duty on boil-up rate, column pressure drop, and overall column efficiency will be examined under total reflux conditions. In the second part, the column will be operated at constant but varying reflux ratios to observe how top and bottom compositions change with time. By analyzing how heat input and reflux ratio influence vapor–liquid equilibrium and column efficiency, this experiment highlights how optimizing these parameters can improve product purity and overall energy efficiency in industrial distillation processes.

2. Experimental

2.1 Apparatus (5 pts)

A couple key pieces of equipment are used in the distillation setup. These include the reboiler, condenser, thermocouples, column, rotameter, pressure differential gauge, valves, and controls. The complete list of equipment, along with images and descriptions, is provided in Table 1.

Table 1 – Laboratory Equipment

Equipment	Number	Picture	Description
Reboiler & Valves	1		Equipment used to reboil the solution and send vapor back into the column. Valves allow drainage and sampling of solution for concentration testing.
Condenser	2		Equipment utilizes a flow of cooling water to cool down vapor to collect ethanol-liquid condensate.

Pressure Differential Gauge	3		Equipment used to measure the drop in pressure across the distillation column.
Rotameter	4		Equipment used to measure and control the flowrate of cooling water entering the condenser to form distillate.

Distillate Valve	5		Valve used to sample distillate from the condenser.
Column & Thermocouples (8)	6		A group of devices used to measure the temperature at varying points of the column.
Control Box	7		Equipment used to control the reboiler duty and reflux ratio of the column.

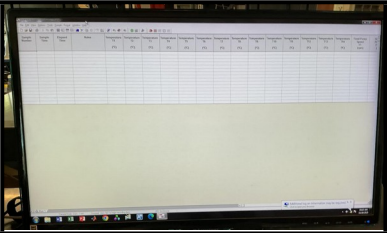
Digital Display	8		Computer records data from the column and tables and figures for easy access.
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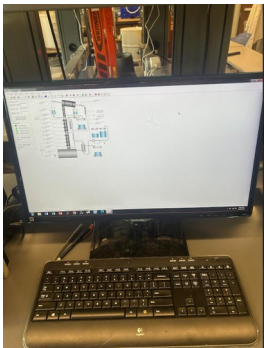
Figure 1: Full Apparatus

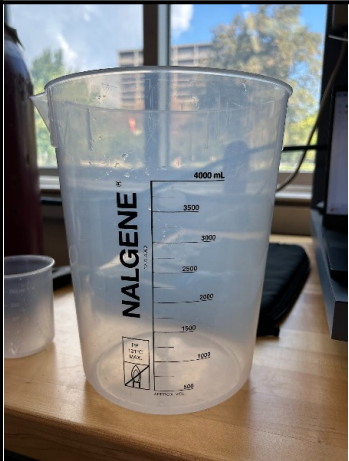
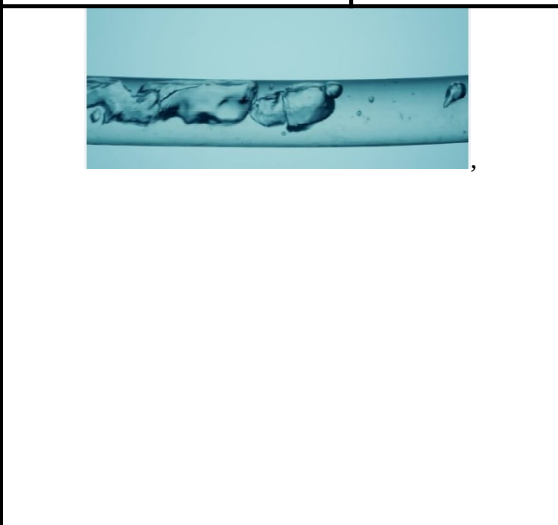


2.2 Materials and Supplies (5 pts)

The complete material and supply list required for operating the fluid flow in pipe systems are provided in Table 2. This table includes all consumables and other resources needed for the experiment, along with relevant details for each item.

Table 2: Material and Supply List

Picture	Number	Description
 Image source: www.payetteforward.com	1	Stopwatch used to time the amount of water being filled in the beaker at a specific time
	2	Used to monitor and record temperature data and control the distillation process through the software interface.

	3	Used to collect and measure liquid volumes during distillation. It allows for accurate measurement of the amount of distilled liquid produced.
	4	Used to measure ethanol concentration in the distilled liquid to check purity and separation efficiency
	5	Test Tube Rack with Test Tubes Holds test tubes for sampling or storing small liquid samples before or after distillation
	7	Used to prepare the ethanol–water mixture for the distillation experiment. Acts as the less volatile component in the mixture being separated.

	8	Funnel Used to safely pour the solution or mixture into the boiler without spilling
	9	Used to protect hands when handling hot glassware or equipment during distillation
	10	Ethanol is used as the volatile component in the ethanol/water mixture to be separated by distillation.

2.3 Experimental Plan (10 pts)

Procedure:

Solution Preparation

-Prepare a 10 L of 10 wt% ethanol/water mixture. From the density of ethanol, $\rho = 0.789$ g/mL, the volume of ethanol in the mixture would be ~ 1.23 L, with the balance being water.

-Load the mixture into the reboiler from the filler cap and close it securely

Experiment A

- 1) Power the control panel. Set the selector to T9 (reboiler). Start condenser cooling water via V5 to ~ 3 L/min.
- 2) Set the reboiler power to 1.50 kW and switch the heater on. Connect the base and top to the manometer (open V6 and V7), then close them after confirming no ΔP that the start.
- 3) Watch the exposed trays (1st and 5th) and note froth, weeping, or entrainment as vapor rises; track T1–T8 as they increase
- 4) When condensate returns to the reflux regulator and the column is flooding each tray uniformly, reduce power to ~ 0.75 kW and allow temperatures T1–T8 to reach steady values.
- 5) Measure the boil-up rate at 0.75 kW by diverting condensate through valve V3 into a cylinder. Discard the first 5–10 mL, then time the collection of a known volume. Keep liquid in the return line to prevent vapor loss.
- 6) At several reboiler power settings (e.g., 0.6–1.2 kW), repeat the boil-up measurement and record the manometer ΔP (open/close V6 & V7 briefly for each reading). Expect ΔP to rise sharply once foaming begins.
- 7) With the column at total reflux and steady state, take representative distillate and bottoms composition samples for later efficiency/plate calculations using the Snap 41 Densitometer. Handle hot bottoms safely.
- 8) Log tray appearance vs. time and power, ΔP vs. boil-up, and all temperatures (T1–T9). These will be used to estimate theoretical plates (Fenske) and column efficiency.

Experiment B

- 1) Use the same ethanol/water charge, set ~ 0.75 kW, and allow total reflux until bubbling is steady on all trays; record temperatures.

- 2) Program a reflux ratio (e.g., 5:1) on the timer: set “back-to-column” duration (C-) and “to receiver” duration (C+), then switch the reflux valve ON; verify condensate to the top receiver.
- 3) Measure the boil-up rate by diverting condensate through valve V3 into a cylinder. Discard 5–10 mL, then time the collection of a known volume. Ensure liquid remains in the return line to prevent vapor loss
- 4) Take paired overhead and bottoms samples nearly simultaneously (overheads via V3; bottoms via V2). Record composition using Snap 41 Densitometer. Keep samples upright to minimize evaporation. Handle bottoms safely
- 5) Repeat step 4 every 5 minutes until you have ≥ 3 paired samples. Record T8 and T1 each time to compute the average column temperature.
- 6) Change to at least one additional reflux ratio (e.g., 3:1 or 10:1) and repeat steps 3–5 to observe how product compositions evolve with R and time.
- 7) After runs, return to total reflux briefly to re-establish steady state before shutdown; then power down and close cooling water.

Variables:

Independent

- Reboiler duty / boil-up rate
- Reflux ratio R
- Time

Controlled

- Feed composition and charge volume
- Stages
- Ambient pressure

Dependent

- Tray/thermocouple temperatures (T1–T9)
- Manometer ΔP across column
- Boil-up rate
- Overhead and bottoms compositions

Data & Model Correlation:

- Use Fenske to estimate the number of theoretical stages from measured top/bottom compositions and average relative volatility, then compute column efficiency
- Plot log–log ΔP vs. boil-up rate; expect a gentle slope at low rates with a sharp rise as foaming/entrainment begins. Use tray observations to annotate regime changes.
- Plot $x_D(t)$ and $x_B(t)$ for each R; higher R should enrich the distillate faster and delay bottoms enrichment. Compare condensate flow totals and compositions over the sampling to the programmed duty cycle.

- ΔP vs. boil-up (log–log) with annotated regime, T1–T9 vs. time to steady state, vs. time for each R, Fenske stage count & efficiency, McCabe–Thiele diagram with data

Project Safety Evaluation (10 pts)

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: The reboiler and control console require electricity. Although the control panel and computer are located safely away from the reboiler, minor risk exists if water is spilled near cords or outlets. Maintain dry hands and surfaces when operating the system.
Potential mechanical hazards?	Yes	If yes, identify conditions: The glass distillation column can crack or shatter if handled roughly. Pressure measurement with the U-tube manometer requires care—opening valves in the wrong order may cause vapor contamination or minor overpressurization. Handle all glassware gently and open valves slowly.
Potential pressure hazards?	No	If yes, identify conditions:
Potential temperature hazards?	Yes	If yes, identify conditions: The reboiler and lower column sections reach temperatures up to ~95 °C. Surfaces may be hot to the touch. Wear heat-resistant gloves when handling or adjusting the reboiler or valves.
Hazardous/toxic chemicals involved?	Yes	If yes, hazards involved: The feed mixture contains approximately 10 vol % ethanol in water. Ethanol is flammable and should be prepared inside a fume hood, away from ignition sources.
Gloves required?	Yes	If yes, specify type: Heat-resistant gloves (when operating column), Nitrile gloves (when making ethanol-water solution)
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Deionized water- SDS Ethanol - SDS
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	About 95 C	The reboiler and column walls can become hot; avoid direct contact and use heat-resistant gloves.
Pressure (psia)	≈ 17-18 psia (slightly above atmospheric)	Operates near atmospheric pressure. Cooling-water loss could cause minor pressure buildup; pressure differences are measured in inches of water
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>

Chemical handling (ethanol mixture)	yes	Prepare the ethanol–water mixture inside a fume hood to minimize vapor exposure and flammability risk. Use nitrile gloves when mixing and transferring the solution.
Hot surface handling	yes	Wear heat-resistant gloves when touching the reboiler, glass column, or valves to prevent burns.
Electrical spill prevention	yes	Avoid spilling liquid near the control console or computer to prevent electrical hazards. Keep cords and surfaces dry during operation.
Pressure measurement	yes	Open manometer valves slowly and in the correct order to prevent vapor contamination or minor overpressurization of the column.
Drain valve management	yes	Ensure the drain valve is closed during operation. Leaving it open can cause ethanol solution loss and improper reboiler function.

I have read relevant background material for the Unit Operations Laboratory entitled: “*Distillation*” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Joshua Olsuji 10/29/2025
signature of team member date

Omar Zaruk 10/29/2025
signature of team member date

Sanad Abu Hweij 10/29/2025
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